

BUSINFO 702 – Group Assignment

Global Patterns in Mental Health Prevalence: Comparing GDP, Health Expenditure and Regional Differences, 2000–2017

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Executive Summary

This project examines how GDP per capita and total health expenditure relate to reported mental health prevalence across countries and regions between 2000 and 2017. An integrated data warehouse was developed using mental health disorder prevalence data from Kaggle, health expenditure per capita from WHO, and GDP per capita from the World Bank.

Three analyses were conducted. First, health spending share generally increased across most regions, while average reported disorder prevalence remained relatively stable. Second, the relationship between GDP per capita and mental health prevalence varied substantially by region, showing positive, negative, and weak associations rather than a consistent global pattern. Third, countries with similar levels of health expenditure could still show persistent disorder-specific differences.

Overall, GDP per capita and total health expenditure alone do not fully explain variation in reported mental health prevalence. For policymakers and health-system planners, the findings highlight the importance of considering resource allocation, healthcare access, and broader social and institutional factors alongside economic indicators when investigating mental health outcomes.

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Project Motivation

Mental health is an important component of sustainable development and is closely connected to broader social and economic conditions (Patel et al., 2018). However, countries differ substantially in their economic capacity, health expenditure, and mental health outcomes. This raises an important question: do differences in national wealth and healthcare resources correspond with differences in mental health outcomes, and are these patterns consistent across regions? Exploring these differences can help reveal where mental health disparities exist and how they vary across countries and over time.

However, investigating these patterns is difficult because relevant information is fragmented across different datasets. This project addresses this gap by developing an integrated data warehouse that creates a consistent country-year view of mental health outcomes, GDP, and health expenditure from 2000–2017, the common period across all three sources. This integration enables comparable descriptive analysis across countries, regions, and time, providing the foundation for examining these patterns through three key questions. Ultimately, this analysis can provide a clearer picture of global disparities and highlight areas that may warrant further investigation.

Research Questions

Mental health is closely connected to broader social and economic conditions. The Lancet Commission highlights the importance of addressing mental health within the context of sustainable development, while also recognizing that mental health outcomes and socioeconomic conditions are interconnected (Patel et al., 2018). However, differences in economic capacity and health spending do not necessarily translate into the same mental health outcomes across countries. Understanding these patterns can help reveal where mental health outcomes differ across regions, how they change over time, and which countries stand out from others with similar resources.

To better understand these differences, we structured our analysis around three key questions:

What is happening across the globe?

This question is important for policymakers because it examines whether changes in general healthcare resource commitment are accompanied by similar changes in reported mental health prevalence across regions.

1. How does health spending share (health expenditure per capita relative to GDP per capita) compare with average mental health disorder occurrence across regions, and how have both measures changed across the 3-year periods spanning 2000 – 2017?

What is the relationship?

This question helps determine whether GDP per capita can be interpreted as a consistent contextual indicator of mental health outcomes, or whether its relationship with prevalence needs to be understood differently across regions.

2. How has the relationship between GDP per capita and mental health disorder rates changed across the 2000–2017 period, and does this pattern differ by region?

Who stands out?

This question helps identify where countries with comparable health spending still show substantially different reported outcomes, providing a basis for prioritizing deeper investigation into resource allocation, service access, diagnosis, and other contextual factors.

3. Among countries with similar levels of health expenditure per capita, which countries show disorder-specific prevalence trends that deviate most from their spending peers across the six 3-year periods from 2000 to 2017?

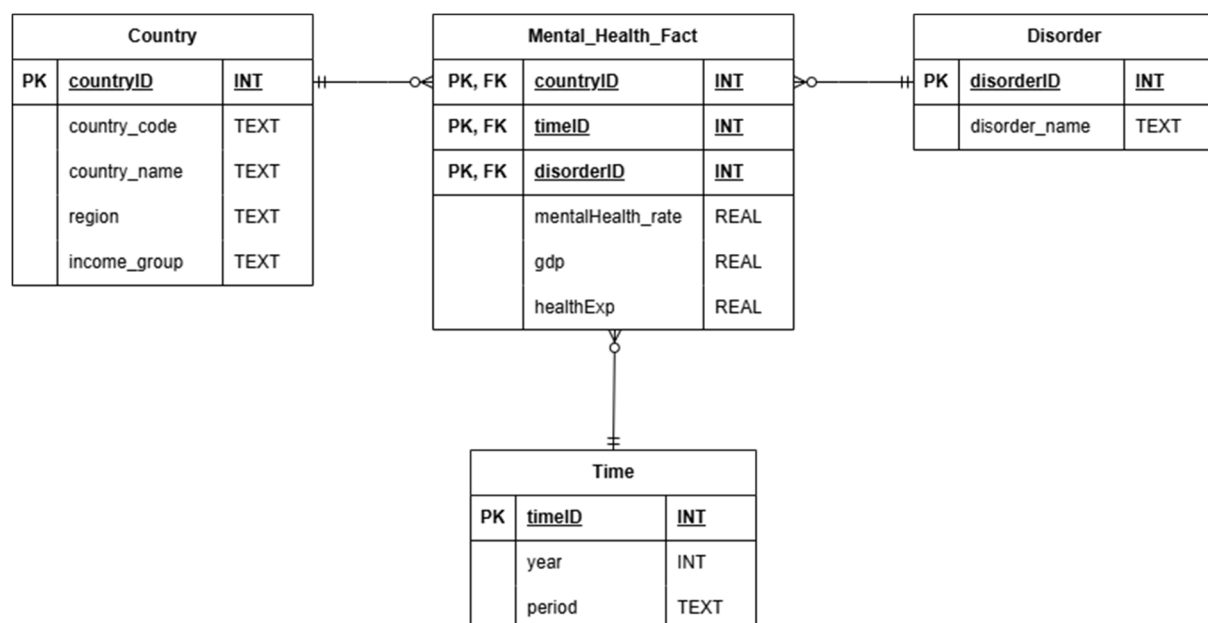
Star Schema Design

The warehouse uses a single fact table (Mental_Health_Fact) at the center, surrounded by three-dimension tables (Country, Time, Disorder) that describe the *who*, *when*, and *what kind* of each fact row.

Fact table grain: one row = one **Country** × **Year** × **Disorder** combination. This is the most important design decision in the schema, it's the finest level of detail any single row represents, and every measure in the fact table must be understood at that grain. Star Schema shown in Figure 1.

Figure 1

Star schema for the Mental Health Data Warehouse



Data Sources and Star Schema

1. Mental Health Disorder, source: Kaggle / Our World in Data

This dataset provides the mental health outcome we want to understand. For each country and year, it tells us the prevalence rate of seven different mental health disorders. This information is used to populate:

- Mental Health Rate in the fact table
- Disorder dimension with the seven disorder types
- Country and Time information used to link the data

2. GDP per Capita – Data Sheet, source: World Bank

This dataset provides information about a country's economic capacity.

- Its GDP per capita values are added to the fact table.

3. GDP per Capita – Metadata Sheet, source: World Bank

This dataset provides information about a country's economic capacity.

- The accompanying World Bank metadata also provides each country's region and income group, which are used in the Country dimension.

4. Current Health Expenditure per Capita, source: WHO

This dataset provides information on healthcare spending in each country.

- The health expenditure per capita value is added to the fact table and can then be compared with mental health outcomes and GDP.

The fact table is supported by three dimensions:

1. Country: country → region, with income group as an additional classification attribute
2. Time: year → 3-year period (attribute hierarchy)
3. Disorder: seven disorder types, with no further hierarchy

The Disorder dimension is intentionally flat, as the seven disorder types do not have a defined parent-child relationship in the source data.

These hierarchies allow the data to be analyzed at different levels of detail. For example, users can drill down from region to individual countries or roll up from individual years into 3-year periods. Income group provides another way to classify and compare countries, but it is not a parent level of the Country hierarchy.

How the Data Fits Together

The fact table combines three measures, mental health prevalence, GDP per capita, and health expenditure, while the dimension tables provide the descriptive information needed to filter, group, and compare the measures.

Because GDP and health expenditure are reported at the country-year level, while mental health data is at the country-year-disorder level, these values are repeated across the seven disorder records for each country-year. This is a deliberate design choice that allows Disorder to remain a queryable dimension while supporting disorder-specific analysis.

ELT Implementation

The warehouse integrates four extracted data sources, which are mental health disorder prevalence, GDP per capita, accompanying country metadata, and current health expenditure per capita, into a star schema implemented in SQLite. The fact table (Mental_Health_Fact) is built at a grain of one row per country, year, and disorder, covering the common analytical timeframe of 2000 to 2017 shared across the three measure-contributing sources.

Data Extraction

All four data sources were extracted directly from their public providers as flat files, requiring no application programming interface (API) access or authentication. No transformation was applied at this stage, files were downloaded in their original structure. Table 1 presents the four extracted data sources used in this project, along with their originating organizations, file formats, and extraction methods.

Table 1

Source Datasets, Formats, and Extraction Methods

Dataset	Source	File Format	Extraction Method
Mental health disorder prevalence	Kaggle	CSV	Direct file download
GDP per capita – Data Sheet	World Bank Open Data	XLSX (exported to CSV)	Direct file download
GDP per capita – Metadata Sheet	World Bank Open Data	XLSX (exported to CSV)	Direct file download
Current health expenditure per capita	WHO Global Health Observatory	CSV	Direct file download

Note. The World Bank workbook includes a second sheet containing country metadata (region and income group classifications), which was extracted alongside the GDP figures because no other source provides this classification at comparable coverage.

Loading

All four extracted files were imported into a SQLite using DB Browser for SQLite's Import CSV wizard, landing each file as an unmodified staging table. No cleaning or restructuring was applied during loading; this step is deferred entirely to the transform stage, consistent with an ELT rather than an ETL architecture. All raw datasets are available in this folder: [Raw Datasets](#)

Import Configuration

Table 2 details the staging table created for each source file, the delimiter applied during import, and the resulting raw row count.

Table 2*SQLite Staging Table Import Configuration*

Staging Table	Source File	Delimiter	Raw Row Count
raw_mental_health	Mental health disorder prevalence CSV	,	108,553
raw_health_expenditure	WHO health expenditure CSV	,	4,585
raw_gdp	World Bank GDP 'Data' sheet (CSV export)	;	265
raw_metadata_gdp	World Bank 'Metadata - Countries' sheet (CSV export)	;	264

Note. The delimiter difference reflects each file's origin: the World Bank workbook exports to semicolon-delimited CSV by regional default, whereas the Kaggle and WHO files are natively comma-delimited. Each import was configured with the matching delimiter in the wizard to ensure columns were parsed correctly, verified visually in the import preview pane prior to confirmation.

Verifying the Load

After import, row counts, missing values, and country coverage were verified directly in SQL before any transformation began, confirming that the staging tables matched the source files exactly and surfacing data-quality issues that the transform stage would later need to resolve.

```
SELECT 'HE' as remarks, Count(*) as total_row -- 4585
FROM raw_health_expenditure
GROUP BY 1
UNION ALL
SELECT 'MH' as remarks, Count(*) as total_row -- 108553
FROM raw_mental_health
GROUP BY 1
UNION ALL
SELECT 'GDP' as remarks, Count(*) as total_row -- 265
FROM raw_gdp
GROUP BY 1
UNION ALL
SELECT 'ME' as remarks , Count (*) as total_row -- 264
FROM raw_metadata_gdp
GROUP BY 1;
```

Row count check across the four fact-contributing sources, confirming that no rows were dropped or duplicated during import.

```
SELECT 'raw mental health' as remarks
, COUNT (DISTINCT code) as total_country --236
FROM raw_mental_health
WHERE code IS NOT NULL AND code != ''
GROUP BY 1
UNION ALL
SELECT 'raw gdp' as remarks
```

```

, COUNT (DISTINCT "Country Code") as total_country -265 (inc. the aggregate
countries)
FROM raw_gdp
GROUP BY 1
UNION ALL
SELECT 'raw health expenditure' as remarks
, COUNT (DISTINCT SpatialDimValueCode) as total_country --195
FROM raw_health_expenditure
WHERE "Location type" = 'Country'
GROUP BY 1
UNION ALL
SELECT 'raw metadata' as remarks
, COUNT (DISTINCT "Country Code") as total_country --217
FROM raw_metadata_gdp
WHERE Region IS NOT NULL AND Region != ''
GROUP BY 1

```

Distinct country-code count per source, establishing each source's coverage prior to integration (236, 265, 217, and 195 countries, respectively).

A missing-value check was then run on the seven disorder percentage columns in `raw_mental_health`, since these columns supply the disorder prevalence measure central to the fact table.

```

-- missing value from mental health
SELECT SUM(CASE WHEN [Schizophrenia (%)] IS NULL THEN 1 ELSE 0 END) AS
null_schizophrenia
, SUM(CASE WHEN [Bipolar disorder (%)] IS NULL THEN 1 ELSE 0 END) AS null_bipolar
, SUM(CASE WHEN [Eating disorders (%)] IS NULL THEN 1 ELSE 0 END) AS null_eating
, SUM(CASE WHEN [Anxiety disorders (%)] IS NULL THEN 1 ELSE 0 END) AS null_anxiety
, SUM(CASE WHEN [Drug use disorders (%)] IS NULL THEN 1 ELSE 0 END) AS null_drug_use
, SUM(CASE WHEN [Depression (%)] IS NULL THEN 1 ELSE 0 END) AS null_depression
, SUM(CASE WHEN [Alcohol use disorders (%)] IS NULL THEN 1 ELSE 0 END) AS
null_alcohol
, COUNT(*) AS total_rows
FROM raw_mental_health;

-- missing value from gdp
SELECT 2000 AS year, SUM(CASE WHEN "2000" IS NULL THEN 1 ELSE 0 END) AS null_count
FROM raw_gdp
UNION ALL
SELECT 2001, SUM(CASE WHEN "2001" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2002, SUM(CASE WHEN "2002" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2003, SUM(CASE WHEN "2003" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2004, SUM(CASE WHEN "2004" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2005, SUM(CASE WHEN "2005" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2006, SUM(CASE WHEN "2006" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2007, SUM(CASE WHEN "2007" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2008, SUM(CASE WHEN "2008" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL

```

```

SELECT 2009, SUM(CASE WHEN "2009" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2010, SUM(CASE WHEN "2010" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2011, SUM(CASE WHEN "2011" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2012, SUM(CASE WHEN "2012" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2013, SUM(CASE WHEN "2013" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2014, SUM(CASE WHEN "2014" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2015, SUM(CASE WHEN "2015" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2016, SUM(CASE WHEN "2016" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
UNION ALL
SELECT 2017, SUM(CASE WHEN "2017" IS NULL THEN 1 ELSE 0 END) FROM raw_gdp
ORDER BY year;

-- missing value from health expenditure
SELECT COUNT (*) as all_rows
, SUM (CASE WHEN FactValueNumeric IS NULL THEN 1 ELSE 0 END) AS missing_rows
FROM raw_health_expenditure;

```

Null counts per disorder column, out of 108,553 raw rows. Individual columns range from 76.2% null (Schizophrenia) to 94.0% null (Anxiety disorders, Drug use disorders, Depression, and Alcohol use disorders).

These null rates are far higher than expected for a genuine prevalence dataset and indicate that `raw_mental_health` concatenates more than one type of record under the same seven-column header. A small subset of rows reports all seven disorders together, while the remainder reports only one or two, with the rest left null. This observation directly motivated the row-filtering logic applied in the transform stage (Step 1 of Fact Table Transformation and Derived Measures), which isolates only the rows where all seven columns are simultaneously populated. By contrast, the equivalent value column in `raw_health_expenditure` (FactValueNumeric) contained no null values, and `raw_gdp`'s year columns were missing only a small, expected number of countries per year (for example, 14 of 265 in 2000), requiring no comparable cleaning step.

A further check was run to anticipate integration gaps ahead of the transform stage, using the mental health dataset as an anchor and left-joining to the other two sources on country code.

```

SELECT DISTINCT m.Entity, m.code, g."Country Code", h.SpatialDimValueCode
FROM raw_mental_health m
LEFT JOIN raw_gdp g ON m.code = g."Country Code"
LEFT JOIN raw_health_expenditure h ON m.code = h.SpatialDimValueCode
WHERE g."Country Code" IS NULL OR h.SpatialDimValueCode IS NULL;

```

This check surfaced 83 entities with incomplete cross-source coverage, caused by mental health aggregate rows (e.g., regional groupings with no country code) and by genuine gaps or aggregate rows in the World Bank and WHO sources. This finding directly informed the join logic used later in the fact table transformation.

Transform

The transform stage builds the star schema entirely in SQL, executed directly against the SQL staging tables. It comprises schema data definition language (DDL), population of the three dimension tables (Country, Time, and Disorder), and population of the fact table (Mental_Health_Fact) via a multi-step common table expression (CTE) pipeline that cleans, unpivots, and joins the three sources.

Star Schema Definition

The schema follows a single-fact star design with a grain of country, year, and disorder. Mental_Health_Fact carries a composite primary key across all three dimension foreign keys, since no single dimension uniquely identifies a fact on its own.

```
DROP TABLE IF EXISTS Mental_Health_Fact;
DROP TABLE IF EXISTS Country;
DROP TABLE IF EXISTS Time;
DROP TABLE IF EXISTS Disorder;

-- Enable foreign key enforcement before creating tables
PRAGMA foreign_keys = ON;

-- Create DIM Country --
CREATE TABLE Country (
    countryID INTEGER PRIMARY KEY AUTOINCREMENT
    , country_code TEXT UNIQUE NOT NULL
    , country_name TEXT NOT NULL
    , region TEXT
    , income_group TEXT
);

-- Create DIM Time --
CREATE TABLE Time (
    timeID INTEGER PRIMARY KEY AUTOINCREMENT
    , year INTEGER UNIQUE NOT NULL
    , period TEXT NOT NULL
);

-- Create DIM Disorder --
CREATE TABLE Disorder (
    disorderID INTEGER PRIMARY KEY AUTOINCREMENT
    , disorder_name TEXT UNIQUE NOT NULL
);

-- Create FACT Mental_Health_Fact --
CREATE TABLE Mental_Health_Fact (
    countryID INTEGER NOT NULL REFERENCES Country(countryID)
    , timeID INTEGER NOT NULL REFERENCES Time(timeID)
    , disorderID INTEGER NOT NULL REFERENCES Disorder(disorderID)
    , mentalHealth_rate REAL
    , gdp REAL
    , healthExp REAL
    , PRIMARY KEY (countryID, timeID, disorderID)
);
```

Full DDL for the three dimension tables and the fact table, including the composite primary key and foreign key constraints enforcing referential integrity between Mental_Health_Fact and each dimension.

Time Dimension Transformation

The Time dimension is generated entirely from a recursive SQL sequence rather than sourced from any raw file, since no dataset provides a standalone calendar table. A CASE expression buckets each year into one of six three-year periods, which serves as the year-to-period attribute hierarchy used throughout the analytical queries.

```
DELETE FROM Time;
INSERT INTO Time (year, period)
SELECT n
      , CASE
          WHEN n BETWEEN 2000 AND 2002 THEN '2000-2002'
          WHEN n BETWEEN 2003 AND 2005 THEN '2003-2005'
          WHEN n BETWEEN 2006 AND 2008 THEN '2006-2008'
          WHEN n BETWEEN 2009 AND 2011 THEN '2009-2011'
          WHEN n BETWEEN 2012 AND 2014 THEN '2012-2014'
          WHEN n BETWEEN 2015 AND 2017 THEN '2015-2017'
        END
FROM (
  WITH RECURSIVE seq(n) AS (
    SELECT 2000
    UNION ALL
    SELECT n + 1 FROM seq WHERE n < 2017
  )
  SELECT n FROM seq
);
```

A recursive common table expression generates the integer sequence from 2000 to 2017; the CASE expression derives the period attribute, giving Time two levels of granularity (year and three-year period) from a single generated column.

Attribute Hierarchies

Two attribute hierarchies are implemented in the warehouse:

1. Time hierarchy: year to period (six three-year buckets, 2000-2002 through 2015-2017), implemented via the CASE expression shown above.
2. Country hierarchy: country to region, populated from the World Bank metadata's region column, enabling analysis to roll up from individual countries to broader geographic regions.

Both hierarchies are used directly by the project's analytical SQL queries to compare trends across periods and regions rather than only at the finest grain.

Other Dimensions: Country and Disorder

The raw_metadata_gdp table was selected as the source for the Country Dimension because it provides the cleanest and most complete country-level data among the available datasets. Although the raw table contains 265 country-code entities, some records represent World Bank aggregate or regional groupings rather than individual countries. By filtering out records with a null or empty Region, the resulting Country Dimension contains 217 valid country entities. The table also provides the required Region and IncomeGroup attributes, making it the most suitable source for the Country Dimension.

```
DELETE FROM Country;
INSERT INTO Country (country_code, country_name, region, income_group)
SELECT DISTINCT [Country Code], TableName, Region, IncomeGroup
FROM raw_metadata_gdp
WHERE Region IS NOT NULL AND Region != '';
```

The *WHERE* clause is the mechanism that separates genuine countries from World Bank aggregate or regional rows in the source file.

The Disorder dimension is populated as a static, controlled list of the seven disorder types tracked in the mental health dataset. Because these are fixed categorical values rather than values derived from row-level data, they are inserted directly rather than extracted via a query.

```
DELETE FROM Disorder;
INSERT INTO Disorder (disorder_name) VALUES
    ('Schizophrenia'), ('Bipolar disorder'), ('Eating disorders'),
    ('Anxiety disorders'), ('Drug use disorders'), ('Depression'),
    ('Alcohol use disorders');
```

Fact Table Transformation and Derived Measures

The fact table is populated by a single INSERT statement built on four common table expressions, each responsible for cleaning or reshaping one part of the source data before the final join. This is the most substantial transformation step, deriving all three fact measures from their respective raw sources.

Step 1: Clean and Isolate Genuine Disorder Prevalence Rows

The raw mental health file concatenates several distinct data blocks under the same column headers, only one of which is genuine disorder prevalence data. Requiring all seven disorder percentage columns to be simultaneously non-null isolates the correct block.

```
WITH
disorder_cleaning as (
SELECT
    Entity AS country_name,
    Code AS country_code,
    Year AS year,
    [Schizophrenia (%)] AS schizophrenia_pct,
    [Bipolar disorder (%)] AS bipolar_pct,
    [Eating disorders (%)] AS eating_disorders_pct,
    [Anxiety disorders (%)] AS anxiety_disorders_pct,
    [Drug use disorders (%)] AS drug_use_disorders_pct,
    [Depression (%)] AS depression_pct,
    [Alcohol use disorders (%)] AS alcohol_use_disorders_pct
FROM raw_mental_health
WHERE [Schizophrenia (%)] IS NOT NULL
    AND [Bipolar disorder (%)] IS NOT NULL
    AND [Eating disorders (%)] IS NOT NULL
    AND [Anxiety disorders (%)] IS NOT NULL
    AND [Drug use disorders (%)] IS NOT NULL
    AND [Depression (%)] IS NOT NULL
    AND [Alcohol use disorders (%)] IS NOT NULL
```

```

AND Code IS NOT NULL AND Code != ''
AND Year BETWEEN 2000 AND 2017
)

```

Step 2: Unpivot Disorder Prevalence (Wide to Long)

SQLite has no native UNPIVOT operator, so the seven wide disorder columns are stacked into long format using UNION ALL, with one SELECT statement per disorder, each aliasing its percentage column to the shared mentalHealth_rate measure. This directly produces the fact table's grain (one row per country, year, and disorder) and derives the mentalHealth_rate measure.

```

, disorder_unpivot as (
SELECT country_code, year, 'Schizophrenia' as disorder_name, schizophrenia_pct
as mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Bipolar disorder' as disorder_name, bipolar_pct as
mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Eating disorders' as disorder_name,
eating_disorders_pct as mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Anxiety disorders' as disorder_name,
anxiety_disorders_pct as mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Drug use disorders' as disorder_name,
drug_use_disorders_pct as mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Depression' as disorder_name, depression_pct as
mentalHealth_rate FROM disorder_cleaning
UNION ALL
SELECT country_code, year, 'Alcohol use disorders' as disorder_name,
alcohol_use_disorders_pct as mentalHealth_rate FROM disorder_cleaning
)

```

Step 3: Clean Health Expenditure Data

These data were filtered to country-level rows only, excluding WHO regional and income aggregates, and restricted to the 2000-2017 timeframe. The period field is explicitly cast from text to integer, which is required for it to join correctly against the Time dimension's integer year column; this step derives the healthExp measure.

```

, health_expenditure_clean as (
SELECT SpatialDimValueCode as country_code
, CAST(Period AS INTEGER) as year
, FactValueNumeric as healthExp
FROM raw_health_expenditure
WHERE Period BETWEEN 2000 AND 2017
AND "Location type" = 'Country'
)

```

Step 4: Unpivot GDP per Capita (Wide to Long)

The same UNION ALL technique used for disorder prevalence is applied to the World Bank GDP data's 18 wide year columns (2000-2017), deriving the GDP measure.

```
,gdp_unpivot as (
  SELECT "Country Name" as country_name, "Country Code" as country_code, 2000 as
  year, "2000" as gdp FROM raw_gdp WHERE "2000" IS NOT NULL
  UNION ALL
  SELECT "Country Name" as country_name, "Country Code" as country_code, 2001 as
  year, "2001" as gdp FROM raw_gdp WHERE "2001" IS NOT NULL
  UNION ALL
  ... (repeated for each year 2002-2016) ...
  UNION ALL
  SELECT "Country Name" as country_name, "Country Code" as country_code, 2017 as
  year, "2017" as gdp FROM raw_gdp WHERE "2017" IS NOT NULL
)
```

Step 5: Combine into the Fact Table

The disorder_unpivot common table expression anchors the grain via an inner join to all three dimensions, so a fact row is created only if its country and disorder both resolve successfully. GDP and health expenditure are brought in via a left join, since their country and year coverage does not fully match the disorder data, consistent with the 83-entity coverage gap identified during loading. This means a fact row can still exist with a null GDP or healthExp value rather than being silently dropped.

```
INSERT INTO Mental_Health_Fact (countryID, timeID, disorderID,
mentalHealth_rate, gdp, healthExp)

WITH
  ... (disorder_cleaning, disorder_unpivot, health_expenditure_clean,
gdp_unpivot CTEs as above) ...

SELECT c.countryID, t.timeID, d.disorderID, du.mentalHealth_rate, g.gdp,
h.healthExp
FROM disorder_unpivot du
JOIN Country c ON c.country_code = du.country_code
JOIN Time t ON t.year = du.year
JOIN Disorder d ON d.disorder_name = du.disorder_name
LEFT JOIN gdp_unpivot g ON g.country_code = du.country_code AND g.year = du.year
LEFT JOIN health_expenditure_clean h ON h.country_code = du.country_code AND
h.year = du.year;
```

The complete common table expression structure is omitted here for brevity; the full, executable version appears in this link: [BUSINFO702_GROUP7_ELT.sql](#)

Validation

After population, the fact table was validated using three checks: total row count, dimensional coverage (countries, disorders, and year range), and source completeness.

```
SELECT Count(*) as total_fact_rows FROM Mental_Health_Fact;

SELECT COUNT(DISTINCT countryID) as total_country
      , COUNT(DISTINCT disorderID) as total_disorder
      , MIN(t.year) as min_year
      , MAX(t.year) as max_year
FROM Mental_Health_Fact m
LEFT JOIN Time t ON m.timeID = t.timeID;

SELECT ROUND(100.0 * SUM(CASE WHEN gdp IS NOT NULL AND healthExp IS NOT NULL
THEN 1 ELSE 0 END) / COUNT(*), 1)
      as pct_rows_all_sources_present
FROM Mental_Health_Fact;
```

Table 3 summarizes the results of these validation checks, confirming that the fact table was populated as intended.

Table 3

Fact Table Validation Results

Check	Result
Total Mental_Health_Fact rows	24,444
Countries represented	194
Disorders represented	7 (all)
Year range	2000–2017
Rows with both gdp and healthExp present	93.5% (22,848 of 24,444)

Note. The 6.5% of rows missing a GDP or health-expenditure value are the expected, explainable consequence of the country-coverage gap identified during loading, and are not a defect introduced during transformation. All figures were produced by executing the complete script against the group's actual source files, confirming that the ELT pipeline runs correctly end-to-end, from raw CSV import through to a fully populated, query-ready fact table.

Final Data Warehouse is available in this link: [BUSINFO702_Group7_Data_Warehouse.db](#)

SQL for Business Analytics

SQL script for answering Research Questions available in this link:

[BUSINFO702_GROUP7_Research_Question_Script.sql](#)

Research Question1:

How does health spending share (health expenditure per capita relative to GDP per capita) compare with average mental health disorder occurrence across regions, and how have both measures changed across the 3-year periods spanning 2000 – 2017?

- **Analysis for RQ1**

Health spending share was calculated as health expenditure per capita divided by GDP per capita and multiplied by 100. Because GDP and health expenditure are recorded at the Country-Year level but repeated across the seven disorder rows in the fact table, the data were first aggregated to one observation per Country-Year to avoid double-counting. The mean of the seven disorder prevalence measures was then used as a descriptive summary of mental health disorder occurrence, and both measures were compared by World Bank region and 3-year period. Only observations with non-missing GDP and health expenditure values were included. The results are shown in Table 4A, 4B, Figure 2, and Figure 3.

- **SQL Logic and Statement Explanation**

```
WITH country_year AS (
  SELECT
    c.country_name,
    c.region,
    t.year,
    t.period,
    m.countryID,
    m.timeID,

    MAX(m.gdp) AS gdp_per_capita,
    MAX(m.healthExp) AS health_exp_per_capita,

    AVG(m.mentalHealth_rate)
      AS avg_mental_health_rate

  FROM Mental_Health_Fact AS m

  JOIN Country AS c
    ON m.countryID = c.countryID

  JOIN Time AS t
    ON m.timeID = t.timeID

  GROUP BY
    c.country_name,
    c.region,
    t.year,
    t.period,
    m.countryID,
    m.timeID
),

complete_cases AS (
  SELECT
```

```

        *,
        100.0 * health_exp_per_capita
        / NULLIF(gdp_per_capita, 0)
        AS health_spending_share
    FROM country_year

    WHERE gdp_per_capita IS NOT NULL
        AND health_exp_per_capita IS NOT NULL
        AND gdp_per_capita > 0
)

SELECT
    region,
    period,

    COUNT(*) AS country_year_observations,

    ROUND(
        AVG(health_spending_share),
        2
    ) AS avg_health_spending_share_pct,

    ROUND(
        AVG(avg_mental_health_rate),
        3
    ) AS avg_mental_health_disorder_rate_pct

FROM complete_cases

GROUP BY
    region,
    period

ORDER BY
    period,
    region;

```

SQL approach: The query was structured using two Common Table Expressions (CTEs) followed by a regional aggregation.

- Create Country-Year observations: Mental_Health_Fact was joined with the Country and Time dimensions. Since GDP and health expenditure are repeated across the seven disorder records for each Country-Year, MAX() was used to retrieve one GDP and health-expenditure value, while AVG() was used to calculate the descriptive mean of the seven mental health disorder prevalence measures.
- Calculate health spending share: A second CTE retained observations with valid GDP and health-expenditure values and calculated health spending share as: Health expenditure per capita / GDP per capita × 100. NULLIF() was used to prevent division by zero.
- Aggregate by region and period: The final query grouped the Country-Year observations by World Bank region and 3-year period. It calculated the average health spending share and average mental health disorder rate for each Region-Period combination.
- Format the output: ROUND() was used to present health spending share to two decimal places and mental health prevalence to three decimal places. The output was ordered by period and region to support comparison over time.

- Result

Table 4A*Average Health Spending Share by Region (%)*

Region	2000–02	2003–05	2006–08	2009–11	2012–14	2015–17
East Asia & Pacific	5.16	5.55	5.66	5.69	5.58	5.73
Europe & Central Asia	6.85	7.44	7.26	7.74	7.70	7.71
Latin America & Caribbean	5.78	5.89	5.89	6.31	6.32	6.53
Middle East & North Africa	5.08	5.08	4.89	5.07	5.17	5.84
North America	10.88	11.76	12.16	13.32	13.13	13.58
South Asia	4.67	4.48	4.61	4.42	4.29	4.48
Sub-Saharan Africa	4.74	5.03	4.95	5.10	5.18	5.28

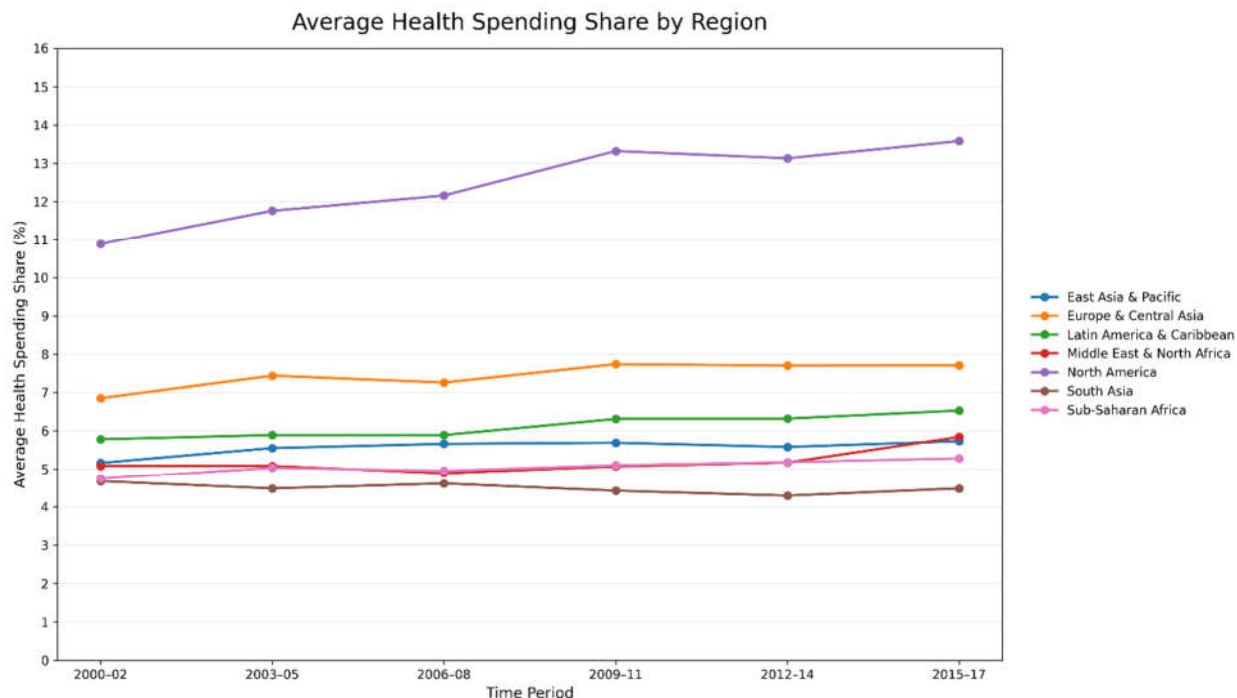
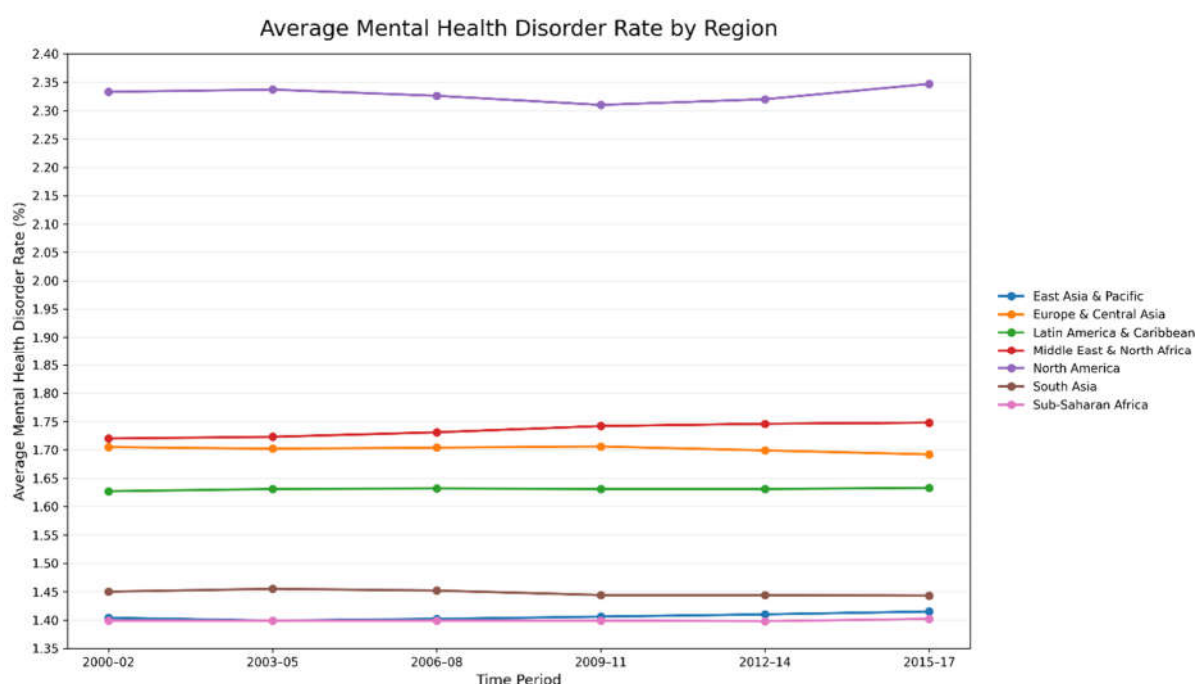
Figure 2*Average Health Spending Share by Region in 3-year periods*

Table 4B*Average Mental Health Disorder Rate by Region (%)*

Region	2000–02	2003–05	2006–08	2009–11	2012–14	2015–17
East Asia & Pacific	1.404	1.399	1.402	1.406	1.410	1.415
Europe & Central Asia	1.705	1.702	1.704	1.706	1.699	1.692
Latin America & Caribbean	1.627	1.631	1.632	1.631	1.631	1.633
Middle East & North Africa	1.720	1.723	1.731	1.742	1.746	1.748
North America	2.333	2.337	2.326	2.310	2.320	2.347
South Asia	1.450	1.455	1.452	1.444	1.444	1.443
Sub-Saharan Africa	1.399	1.399	1.399	1.399	1.398	1.402

Figure 3*Average Mental Health Disorder Rate by Region*

- Interpretation**

The results show that health spending share generally increased across most regions between 2000 and 2017, whereas average reported mental health disorder occurrence remained comparatively stable. North America recorded both the highest health spending share and the largest increase, rising from 10.88% in 2000–2002 to 13.58% in 2015–2017. Europe and Central Asia increased from 6.85% to 7.71%, while Latin America and the Caribbean increased from 5.78% to 6.53%. South Asia was the

only region whose health spending share ended below its initial level, declining slightly from 4.67% to 4.48%.

In contrast, changes in average mental health disorder prevalence were small across all regions. Europe and Central Asia decreased slightly from 1.705% to 1.692%, while South Asia declined from 1.450% to 1.443%. The Middle East and North Africa showed the largest increase in the descriptive mental health measure, from 1.720% to 1.748%, although the absolute change was still small. North America maintained the highest average reported disorder prevalence, increasing slightly from 2.333% to 2.347%, despite also having the highest health spending share.

Overall, no consistent inverse pattern was observed in which a higher health spending share corresponded to lower reported mental health disorder prevalence. Most regions increased their relative healthcare spending while mental health prevalence changed only marginally or, in some cases, increased. These results describe an observed pattern between two national-level measures and should not be read as evidence about the effect of mental-health-specific spending in particular (see note above). Other socioeconomic, healthcare-access, demographic, reporting, and country-specific factors not captured in this warehouse may also be associated with the observed patterns.

These results represent descriptive associations rather than causal relationships. In addition, regional sample sizes are unequal; for example, North America contains only six complete Country-Year observations in each 3-year period, so its regional estimates should be interpreted cautiously. The warehouse also contains some missing GDP and health expenditure values, which is why RQ1 uses complete cases for the required economic measures.

From a health-policy perspective, this means that total healthcare spending share should not be used alone as a benchmark for mental health performance. Regions with increasing expenditure but little corresponding change in reported prevalence may warrant closer examination of how resources are allocated, whether mental health services are accessible, and how effectively broader health-system resources reach mental health needs.

Research Question 2:

How has the relationship between GDP per capita and mental health disorder rates changed across the 2000–2017 period, and does this pattern differ by region?

- **Analysis for RQ2**

To answer RQ2, the mean of the seven disorder prevalence measures was used as a descriptive summary of mental health disorder occurrence for each Country-Year. These observations were then averaged within each 3-year period to create one record per Country-Period. Pearson's correlation coefficient was calculated between GDP per capita and the mean disorder prevalence separately for each World Bank region and 3-year period. This allows both changes over time and differences in the GDP–mental health relationship across regions to be compared. Results are shown in Table 5 and Figure 4.

- **SQL Logic and Statement Explanation**

```

WITH country_year AS (
    SELECT
        c.countryID,
        c.country_name,
        c.region,
        t.period,
        t.year,
        MAX(m.gdp) AS gdp_per_capita,
        AVG(m.mentalHealth_rate) AS avg_mental_health_rate
    FROM Mental_Health_Fact AS m
    JOIN Country AS c
        ON m.countryID = c.countryID
    JOIN Time AS t
        ON m.timeID = t.timeID
    WHERE m.gdp IS NOT NULL
        AND m.gdp > 0
    GROUP BY
        c.countryID,
        c.country_name,
        c.region,
        t.period,
        t.year
),

country_period AS (
    SELECT
        countryID,
        country_name,
        region,
        period,
        AVG(gdp_per_capita) AS avg_gdp_per_capita,
        AVG(avg_mental_health_rate) AS avg_mental_health_rate
    FROM country_year
    GROUP BY
        countryID,
        country_name,
        region,
        period
),

correlation_stats AS (
    SELECT
        region,
        period,
        COUNT(*) AS n,
        SUM(avg_gdp_per_capita) AS sum_x,
        SUM(avg_mental_health_rate) AS sum_y,
        SUM(avg_gdp_per_capita * avg_mental_health_rate) AS sum_xy,
        SUM(avg_gdp_per_capita * avg_gdp_per_capita) AS sum_x2,
        SUM(avg_mental_health_rate * avg_mental_health_rate) AS sum_y2
    FROM country_period
    GROUP BY
        region,
        period
)

SELECT
    region,
    period,
    n AS countries,
    ROUND(
        (n * sum_xy - sum_x * sum_y)
        /
        NULLIF(
            SQRT(
                (n * sum_x2 - sum_x * sum_x)

```

```

        *
        (n * sum_y2 - sum_y * sum_y)
    ),
    0
),
3
) AS pearson_r
FROM correlation_stats
ORDER BY
    region,
    period;

```

SQL approach: The RQ2 query used three CTEs followed by a final calculation of Pearson's correlation coefficient.

- Create Country-Year observations: The fact table was joined with the Country and Time dimensions. GDP was reduced to one value per Country-Year using MAX(), while the seven disorder prevalence measures were summarised using AVG(). Records with missing or non-positive GDP values were excluded.
- Aggregate to Country-Period level: Country-Year observations were averaged within each 3-year period. This generated one GDP value and one descriptive mental health prevalence value for each Country-Period.
- Prepare correlation components: For each Region-Period group, the query calculated the number of countries and all values required for the Pearson correlation formula, including the sums of GDP, mental health prevalence, their cross-products, and their squared values.
- Calculate Pearson's correlation: The final SELECT applied the Pearson correlation formula to each Region-Period group. NULLIF() was used to avoid division by zero where variation was insufficient, and the resulting coefficient was rounded to three decimal places.
- Interpret the output: Pearson's r ranges from -1 to +1. Positive values indicate that higher GDP is associated with higher reported prevalence within the Region-Period, negative values indicate an inverse association, and values close to zero indicate little linear association.

• Result

Table 5

Pearson Correlation Between GDP per Capita and Average Mental Health Disorder Prevalence by Region and 3-Year Period

Region	Countries	2000– 2002	2003– 2005	2006– 2008	2009– 2011	2012– 2014	2015– 2017
East Asia & Pacific	29	0.419	0.558	0.595	0.610	0.665	0.645
Europe & Central Asia	50	0.632	0.641	0.630	0.616	0.616	0.629
Latin America & Caribbean	34	0.147	0.147	0.202	0.259	0.302	0.282

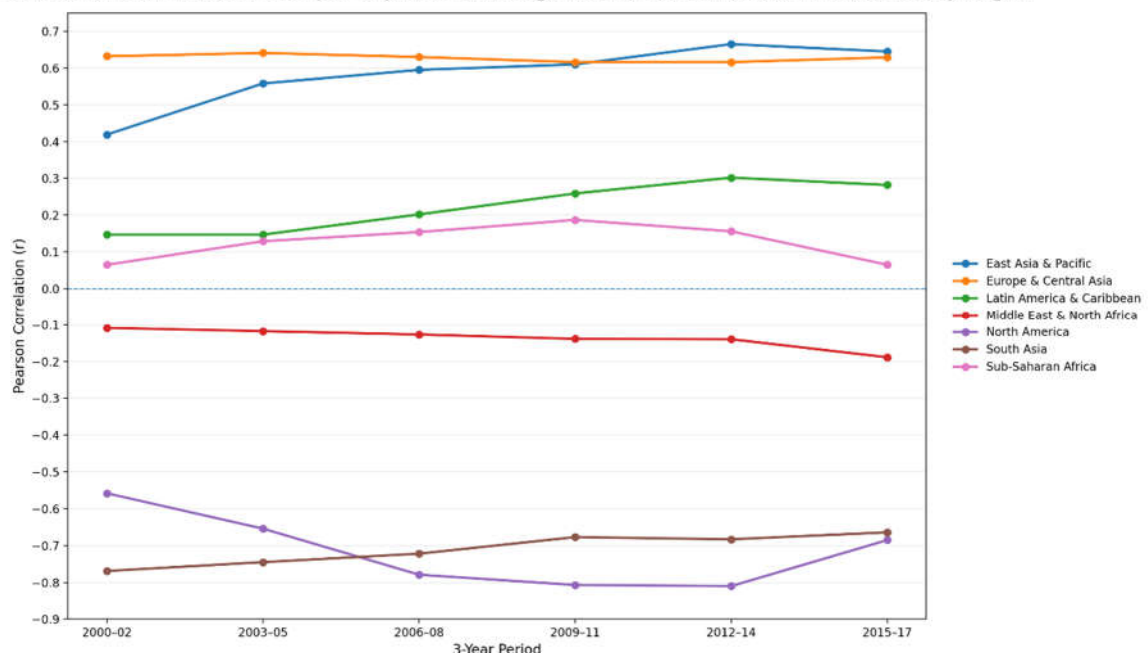
Middle East & North Africa	23	-0.109	-0.118	-0.127	-0.139	-0.140	-0.189
North America	3	-0.558	-0.654	-0.779	-0.807	-0.810	-0.685
South Asia	6	-0.769	-0.745	-0.722	-0.677	-0.683	-0.664
Sub-Saharan Africa	47–48	0.065	0.129	0.154	0.187	0.156	0.065

Note. Average mental health disorder prevalence represents the mean of the seven selected disorder prevalence indicators. Pearson's r was calculated across countries within each region and 3-year period.

Figure 4

Pearson correlation between GDP and the average rate of mental health disorders by region.

Pearson Correlation Between GDP per Capita and Average Mental Health Disorder Prevalence by Region



- Interpretation**

The results indicate that the relationship between GDP per capita and reported mental health disorder prevalence varied substantially across regions. Europe and Central Asia showed a consistently positive relationship throughout the study period, with Pearson correlation coefficients remaining between approximately 0.62 and 0.64. East Asia and the Pacific also displayed a positive relationship, which strengthened from 0.419 in 2000–2002 to 0.645 in 2015–2017. Latin America and the Caribbean showed a weaker positive association, although the correlation generally increased over time.

In contrast, South Asia exhibited a consistently negative relationship between GDP per capita and average reported disorder prevalence. The correlation changed from

−0.769 in 2000–2002 to −0.664 in 2015–2017, indicating that the negative relationship weakened slightly but remained substantial. The Middle East and North Africa also showed negative correlations, although these remained relatively weak. Sub-Saharan Africa had correlations close to zero across all periods, suggesting little consistent linear association between GDP per capita and the descriptive mental health measure.

North America also showed relatively strong negative correlations, reaching −0.810 in 2012–2014. However, this result should be interpreted cautiously because the regional analysis includes only three countries, making the correlation sensitive to individual-country differences.

Overall, the direction and strength of the association between GDP per capita and reported mental health prevalence differed markedly by region rather than following one consistent pattern, positive in some regions, negative in others, and near zero elsewhere. This describes how the association varies by region; it is not evidence that national economic capacity determines reported mental health outcomes. Other factors not captured in this warehouse, including healthcare access, diagnosis and reporting practices, demographic characteristics, and broader social conditions, may also be associated with the observed patterns. These correlations represent associations and should not be interpreted as causal effects.

For international organisations and policymakers, this regional variation means that GDP per capita should not be treated as a universal proxy for mental health performance. Economic and mental health indicators should instead be interpreted within regional and institutional contexts, particularly where small regional samples or different diagnostic and reporting systems may affect the observed relationship.

Research Question 3:

Among countries with similar levels of health expenditure per capita, which countries show disorder-specific prevalence trends that deviate most from their spending peers across the six 3-year periods from 2000 to 2017?

- **Analysis for RQ3**

To answer RQ3, countries were grouped into four health-expenditure quartiles for each 3-year period based on their average per capita health expenditure. Countries within the same quartile were treated as spending peers. For each disorder and period, the average prevalence of each country was compared with the mean prevalence of countries in the same spending quartile using a z-score. The sequence of these period-level deviations was used to represent each country's prevalence pattern across the six periods. Countries were then ranked by their average absolute z-score across all six periods, with larger values indicating greater average deviation from countries with comparable healthcare spending. Persistence was assessed separately using the number of periods in which the absolute z-score was at least 2. An absolute z-score of 2 or greater ($|z| \geq 2$) was defined as a notable deviation, and outlier periods records the number of 3-year periods in which this threshold was exceeded. The result is shown in Table 6 and Figure 5.

- **SQL Logic and Statement Explanation**

```

WITH country_year AS (
    SELECT
        c.countryID,
        c.country_name,
        t.year,
        t.period,
        MAX(m.healthExp) AS health_exp_per_capita
    FROM Mental_Health_Fact AS m
    JOIN Country AS c
        ON m.countryID = c.countryID
    JOIN Time AS t
        ON m.timeID = t.timeID
    WHERE m.healthExp IS NOT NULL
    GROUP BY
        c.countryID,
        c.country_name,
        t.year,
        t.period
),

country_period_spending AS (
    SELECT
        countryID,
        country_name,
        period,
        AVG(health_exp_per_capita)
        AS avg_health_exp_per_capita
    FROM country_year
    GROUP BY
        countryID,
        country_name,
        period
),

spending_groups AS (
    SELECT
        *,
        NTILE(4) OVER (
            PARTITION BY period
            ORDER BY avg_health_exp_per_capita
        ) AS spending_quartile
    FROM country_period_spending
),

country_period_disorder AS (
    SELECT
        c.countryID,
        c.country_name,
        t.period,
        d.disorder_name,
        AVG(m.mentalHealth_rate)
        AS avg_prevalence
    FROM Mental_Health_Fact AS m
    JOIN Country AS c
        ON m.countryID = c.countryID
    JOIN Time AS t
        ON m.timeID = t.timeID
    JOIN Disorder AS d
        ON m.disorderID = d.disorderID
    GROUP BY
        c.countryID,
        c.country_name,
        t.period,
        d.disorder_name
),

```

```

peer_data AS (
    SELECT
        cpd.countryID,
        cpd.country_name,
        cpd.period,
        cpd.disorder_name,
        cpd.avg_prevalence,
        sg.avg_health_exp_per_capita,
        sg.spending_quartile
    FROM country_period_disorder AS cpd
    JOIN spending_groups AS sg
        ON cpd.countryID = sg.countryID
        AND cpd.period = sg.period
),

peer_stats AS (
    SELECT
        *,
        AVG(avg_prevalence) OVER (
            PARTITION BY
                period,
                spending_quartile,
                disorder_name
        ) AS peer_mean,

        AVG(avg_prevalence * avg_prevalence) OVER (
            PARTITION BY
                period,
                spending_quartile,
                disorder_name
        ) AS peer_mean_sq

    FROM peer_data
),

scored AS (
    SELECT
        *,
        CASE
            WHEN peer_mean_sq - peer_mean * peer_mean > 0
            THEN
                (avg_prevalence - peer_mean)
                /
                SQRT(
                    peer_mean_sq
                    - peer_mean * peer_mean
                )
            END AS z_score
    FROM peer_stats
),

summary AS (
    SELECT
        country_name,
        disorder_name,
        COUNT(*) AS periods_observed,

        ROUND(
            AVG(ABS(z_score)), 2
        ) AS avg_abs_z,

        ROUND(
            AVG(z_score), 2
        ) AS avg_signed_z,

        ROUND(
            MAX(ABS(z_score)), 2
        ) AS max_abs_z,

```

```

        SUM(
            CASE
                WHEN ABS(z_score) >= 2
                THEN 1
                ELSE 0
            END
        ) AS outlier_periods

FROM scored

GROUP BY
    country_name,
    disorder_name

HAVING COUNT(*) = 6
)

SELECT
    country_name,
    disorder_name,
    avg_abs_z,
    avg_signed_z,
    max_abs_z,
    outlier_periods
FROM summary

ORDER BY
    avg_abs_z DESC,
    country_name,
    disorder_name

LIMIT 10;

```

SQL approach: The RQ3 query used a sequence of CTEs and window functions to construct health-expenditure peer groups and identify persistent disorder-specific deviations.

- Calculate Country-Period health expenditure: Health expenditure was first reduced to one value per Country-Year and then averaged across each 3-year period, producing an average health expenditure per capita for each Country-Period.
- Create spending-peer groups: NTILE (4) was applied separately within each 3-year period to divide countries into four health-expenditure quartiles. Countries within the same quartile and period were treated as spending peers.
- Calculate disorder-specific prevalence: Mental health observations were aggregated to the Country-Period-Disorder level, giving one average prevalence value for each disorder, country and 3-year period.
- Match countries with their spending peers: The expenditure-quartile data were joined with the disorder-prevalence data so that every Country-Period-Disorder observation was associated with its corresponding spending group.
- Calculate peer-group statistics: SQL window functions calculated the mean prevalence and variance for each combination of 3-year period, spending quartile and disorder type.
- Standardise deviations using z-scores: Each country's disorder prevalence was converted into a z-score relative to its spending-peer group. Positive z-scores

indicate prevalence above the peer mean, while negative values indicate prevalence below the peer mean.

- Measure magnitude and persistence: For each Country-Disorder combination, the query calculated the average absolute z-score, average signed z-score, and maximum absolute z-score to measure the magnitude and direction of deviation. Persistence was assessed by counting the number of periods in which the absolute z-score was at least 2. Only combinations observed across all six 3-year periods were retained.
- Rank the largest deviations: Finally, Country-Disorder combinations were ranked by average absolute z-score, and the ten largest, persistent deviations were returned.

• Result

The analysis identified the ten country-disorder combinations with the largest average absolute deviations from countries in comparable health-expenditure quartiles. All ten combinations exceeded the notable-deviation threshold of $|z| \geq 2$ in all six 3-year periods, indicating persistent rather than temporary differences from their spending peers.

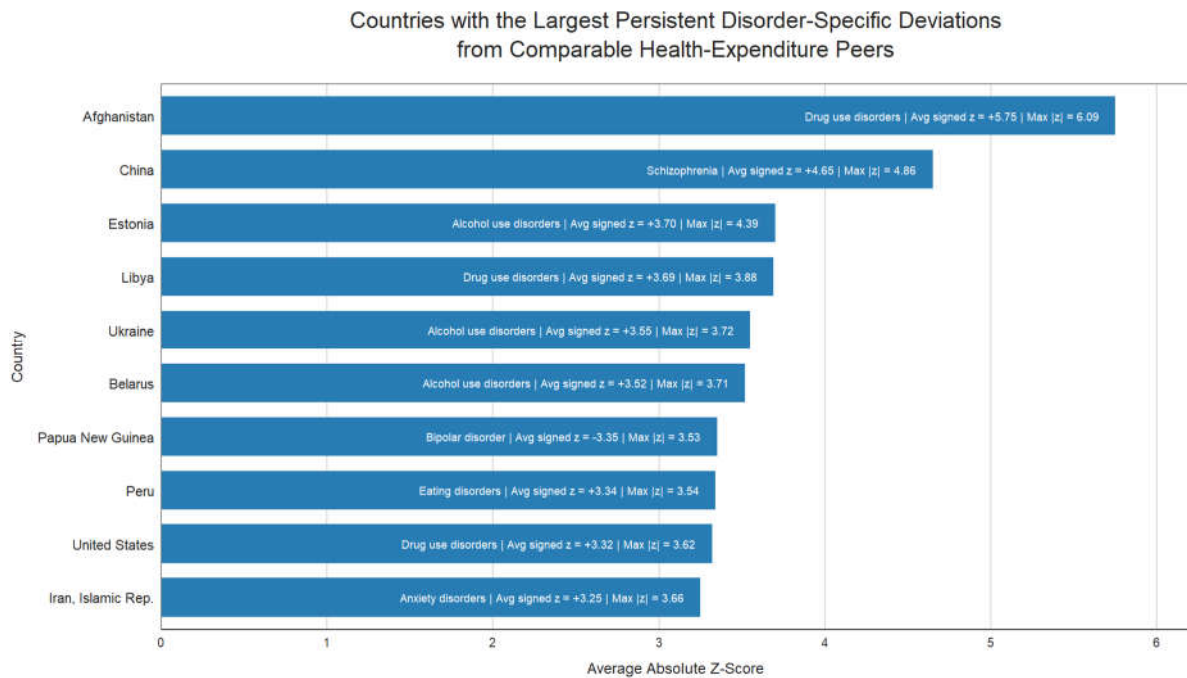
Table 6

Countries with the Largest Persistent Disorder-Specific Deviations from Comparable Health-Expenditure Peers.

Country Name	Disorder	avg_abs_z	avg_signed_z	max_abs_z	Outlying Periods
Afghanistan	Drug use disorders	5.75	5.75	6.09	6
China	Schizophrenia	4.65	4.65	4.86	6
Estonia	Alcohol use disorders	3.7	3.7	4.39	6
Libya	Drug use disorders	3.69	3.69	3.88	6
Ukraine	Alcohol use disorders	3.55	3.55	3.72	6
Belarus	Alcohol use disorders	3.52	3.52	3.71	6
Papua New Guinea	Bipolar disorder	3.35	-3.35	3.53	6
Peru	Eating disorders	3.34	3.34	3.54	6
United States	Drug use disorders	3.32	3.32	3.62	6
Iran, Islamic Rep.	Anxiety disorders	3.25	3.25	3.66	6

Figure 5

Countries that deviate the most from their comparable Health-Expenditure peers.



- Interpretation**

The results reveal substantial and persistent differences in disorder-specific prevalence among countries with comparable levels of health expenditure per capita. Afghanistan showed the largest deviation, with drug use disorder prevalence averaging 5.75 standard deviations above the mean of its spending peers across the six 3-year periods. China also recorded a large and persistent positive deviation in schizophrenia prevalence, with an average z-score of 4.65. Estonia, Libya, Ukraine, and Belarus similarly showed consistently elevated prevalence for particular disorders relative to countries in comparable expenditure quartiles.

Most of the largest deviations were positive, indicating prevalence above the relevant spending-peer average. However, Papua New Guinea represented an important exception: its bipolar disorder prevalence had an average signed z-score of -3.35 , indicating that it remained substantially below the mean of countries with similar levels of health expenditure.

Importantly, all ten of the highest-ranked country-disorder combinations exceeded the $|z| \geq 2$ threshold in all six 3-year periods. This suggests the observed differences were persistent across the 2000–2017 period rather than being driven by isolated years or temporary fluctuations.

Overall, the findings show that comparable levels of total health expenditure per capita do not correspond to comparable reported mental health outcomes across countries. This describes a pattern of divergence among spending peers; it is not evidence about what specifically drives that divergence and should not be read as a statement about mental-health-specific funding, which this warehouse does not

measure. Other factors not captured here, such as healthcare allocation, service accessibility, diagnostic and reporting practices, demographic characteristics, and broader social conditions, may also be associated with the observed patterns. These deviations should be interpreted as indicators for further investigation rather than evidence of causal effects.

For health-system planners, these persistent outliers provide a practical screening tool to identify countries where service delivery, resource allocation, healthcare access, or broader social conditions may warrant more targeted investigation.

Insight, Future Work and Limitations

Key Insights and Stakeholder Implications

The analysis shows that higher economic capacity or greater total health expenditure does not consistently correspond with lower reported mental health prevalence. Although health spending share increased across most regions between 2000 and 2017, average disorder prevalence remained relatively stable. For policymakers and health-system planners, this means that total spending alone is an incomplete benchmark for mental health performance. Where spending increases without corresponding changes in reported prevalence, the allocation, accessibility, and delivery of mental health services may warrant closer investigation.

A second insight is that the GDP–mental health relationship is strongly context-dependent. The association was positive in some regions, negative in others, and weak in Sub-Saharan Africa. This matters because a single global assumption that greater national wealth corresponds with better mental health outcomes would be misleading. International organisations and policymakers should therefore interpret economic and mental health indicators within regional and institutional contexts rather than relying on GDP as a universal proxy for mental health performance.

The spending-peer analysis provides a further practical insight: countries with similar levels of total health expenditure can still show persistent disorder-specific differences. These outliers can be used as screening signals rather than performance rankings. Countries that repeatedly diverge from comparable spending peers may merit targeted investigation into service delivery, resource allocation, healthcare access, diagnostic practices, or wider social conditions.

Taken together, the findings show that GDP and total healthcare expenditure describe only part of the variation in reported mental health outcomes. The practical value of the warehouse is therefore not to claim that economic resources determine mental health, but to identify where expected relationships do not hold and where deeper investigation can be prioritised.

Future Work

The highest-priority next step is to incorporate mental-health-specific expenditure and health-system capacity indicators. The current WHO measure captures total healthcare spending and therefore cannot show how much funding actually reaches mental health services. Adding mental health expenditure, psychiatric workforce, treatment coverage, and service accessibility would allow future analysis to investigate whether persistent differences between spending peers reflect differences in resource allocation or service capacity. This directly addresses the main interpretive limitation of the current project and would make the findings more actionable for health-system planners.

A second priority is to explain the regional and country-level variation identified in RQ2 and RQ3. Variables such as unemployment, income inequality, education, population

structure, and other social conditions could be integrated into the warehouse. These measures are important because GDP and total health expenditure alone did not explain the contrasting regional relationships or persistent country-level outliers. Their inclusion would help determine which contextual factors are most closely associated with the remaining differences.

Future analysis should then move from bivariate comparisons towards multivariable and longitudinal methods. Panel-data or multivariable models could test whether GDP, health expenditure, and health-system indicators remain associated with mental health prevalence after accounting for country characteristics and changes over time. This would provide stronger evidence about which factors are most informative, although the observational data would still not establish causality.

Comparability could also be improved by using purchasing-power-parity or inflation-adjusted economic measures and population-weighted regional estimates. These changes would reduce distortions arising from price-level differences and unequal country population sizes, improving the reliability of comparisons across regions and periods.

Limitations

- Health expenditure per capita measures overall health system investment, not mental-health-specific spending, per the WHO Mental Health Atlas 2020, mental health receives a global median of only 2.1% of government health budgets, regardless of total health spending (World Health Organization, 2021), so RQ1 findings should be read as reflecting general health system capacity rather than targeted mental health investment.
- 93.5% of fact rows have both GDP and health expenditure present, the remaining 6.5% (linked to 83 entities with incomplete source coverage) were retained with NULL values rather than discarded, to avoid systematically excluding countries with less consistent international reporting.
- Regional averages in RQ1 and RQ2 are unweighted averages of country-level values, meaning a small country contributes equally to the regional average as a large one; this was a deliberate choice for comparative purposes but means results do not reflect population-weighted regional totals.
- Findings are descriptive and correlational. The star-schema/SQL approach used here identifies patterns and group differences; it does not establish causal relationships between wealth, health expenditure, and disorder occurrence.

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